

■ ARASH RAHMANIDEHKORDI

Email: arahmani@charlotte.edu || GitHub: rdehkordi-arash.github.io

Scholar: scholar.google.com/citations?user=TD0X9Z0AAAAJhl=enauthuser=loi=ao

Department of Mechanical Engineering, University of North Carolina at Charlotte, NC 28223, USA

RESEARCH BACKGROUND

Research expertise in nonlinear systems, Model Predictive Control (MPC), and robust control, with emphasis on Feedback Linearization (FL/FFL). Current research focuses on hybrid FL-MPC architectures for uncertain and constrained systems, mixed-autonomy highway traffic control, and learning-based methods including Gaussian Processes (GP) and Belief-Space MPC.

RESEARCH INTERESTS

- Nonlinear Systems
- Robust and Constrained Control
- Robotics & Autonomous Systems
- Traffic Control
- Intelligent Transportation Systems
- Data-driven Control

EDUCATION

Ph.D. in Mechanical Engineering (Dynamics & Control, GPA:3.85) 2023 - Present
University of North Carolina at Charlotte, Charlotte, NC, USA

Thesis: *"Design and Implementation of a Constraint-Aware Controller for Nonlinear and Uncertain Systems: A Traffic Density Management Application"*

M.Sc. in Mechatronics 2022
University of Isfahan, Isfahan, Iran

B.Sc. in Mechanical Engineering 2018
Yazd University, Yazd, Iran

PUBLICATIONS

- JOURNAL

- [1] **Arash Rahmanidehkordi**, Amir H. Ghasemi, "Traffic density control for heterogeneous highway systems with input constraints," *IEEE Control Systems Letters*, vol. 8, pp. 2787–2792, 2024.
 - [2] **Arash Rahmanidehkordi**, Amirhossein Ghasemi, "Filtered dynamic inversion for input constrained uncertain linear systems with unknown and unmeasured disturbances," *International Journal of Dynamics and Control*, vol. 14, no. 8, p. 286, 2026. DOI: 10.1007/s40435-026-02224-9.
 - [3] Pouria Karimi Shahri, **Arash Rahmanidehkordi**, Azad Ghaffari, Amir H. Ghasemi, "Enhancing traffic flow in heterogeneous freeways: Integration of multivariable extremum seeking and filtered feedback linearization control," *IEEE Access*, vol. 13, pp. 129 573–129 587, 2025. DOI: 10.1109/ACCESS.2025.3585431.
 - [4] Pouria Karimi Shahri, **Arash Rahmanidehkordi**, Azad Ghaffari, Amir H. Ghasemi, "Control of heterogeneous freeways using newton-based extremum seeking and filtered feedback linearization," *ASME Letters in Dynamic Systems and Control*, 2026 (Accepted).
 - [5] **Arash Rahmanidehkordi**, Amir H. Ghasemi, "Input-constrained filtered feedback linearization for control-affine nonlinear systems subject to unknown disturbances," *International Journal of Robust and Nonlinear Control*, 2026 (Submitted).
 - [6] **Arash Rahmanidehkordi**, Amirhossein Ghasemi, "Enhancing traffic flow via feedback linearization and model predictive control under input constraints," *IFAC-PapersOnLine*, vol. 58, no. 28, pp. 372–377, 2024.
 - [7] **Arash Rahmanidehkordi**, Amir H Ghasemi, "Traffic density control via filtered feedback linearization in homogeneous highway corridors with input constraints," *American Control Conference*, pp. 4125–4131, 2026.

Reviewer: IEEE Control Systems Letters (L-CSS); International Journal of Dynamics and Control; Scientific Reports; Conference on Decision and Control (CDC); American Control Conference (ACC);

PRESENTATIONS

- “Enhancing Traffic Flow via Feedback Linearization and Model Predictive Control under Input Constraints.” Paper presentation at the *Modeling, Estimation, and Control Conference (MECC)*, Chicago, IL, October 29, 2024.
- “Traffic Density Control for Heterogeneous Highway Systems with Input Constraints.” Paper presentation at the *American Control Conference (ACC)*, Denver, CO, July 9, 2025.
- “Traffic Density Control via Filtered Feedback Linearization in Homogeneous Highway Corridors with Input Constraints.” Paper presentation at the *American Control Conference (ACC)*, New Orleans, LA, May 29, 2026.

AWARDS & FELLOWSHIP

- **Graduate Student Award – Distinguished Researcher**, Department of Mechanical Engineering and Engineering Science, University of North Carolina at Charlotte 2025
- **National Science Foundation (NSF) Grant (#2130704)**, Graduate Research Assistant, “*Learning-Based Scalable Predictive Control Strategies for Heterogeneous Traffic Networks*”
- **National Science Foundation (NSF) Carrier Award, Grant (#2238268)**, Graduate Research Assistant, “*Learning and Leveraging Conventions in the Design of an Adaptive Haptic Shared Control for Steering a Semi-Automated Vehicle*”
- **Ranked 1st**, M.Sc. in Mechatronics Class of 2019, Isfahan University, Isfahan, Iran 2021

RESEARCH EXPERIENCE

Hybrid Feedback Linearization (FL)–MPC for Heterogeneous Traffic Flow Control

Developed a traffic management hybrid controller for highway corridors with mixed traffic (human-driven and autonomous vehicles) based on the heterogeneous METANET model. Formulated the traffic flow as an over-actuated, nonlinear system. Designed a hybrid FL-MPC controller to regulate vehicle densities under input constraints to reduce congestion time. Implemented a novel constraint-mapping algorithm within the MPC layer to translate virtual control limits into feasible physical commands.

key words: Traffic Control, Feedback Linearization, Constraint Awareness

Filtered Feedback Linearization for Input-Constrained Systems under Disturbances

Developed a novel control architecture integrating FFL with MPC for nonlinear systems subject to unknown disturbances and actuator constraints. Leveraged FFL’s inherent robustness by passing the input through a low-pass filter while embedding MPC to ensure constraint satisfaction. Formulated a novel constraint-mapping algorithm that dynamically translates actuator bounds into the virtual control domain, preserving stability and the low-pass filtering properties of FFL.

key words: Nonlinear Systems, Filtered Feedback Linearization, Robust Control, MPC

Sensor Fusion and Motion Planning for Autonomous Vehicle Navigation

Developed the control framework for autonomous vehicles by integrating data from GPS, LiDAR, and IMU sensors operating at different sampling rates. Implemented an Extended Kalman Filter (EKF) for multi-sensor fusion, using high-rate IMU data for short-term prediction and periodic GPS/LiDAR updates for global correction, achieving robust state estimation and drift reduction. Designed a motion-planning and trajectory-tracking controller for vehicle path following under sensor noise.

key words: Extended kalman filter (EKF), Sensor fusion, Motion planning

Delta Robot Kinematic and Dynamic Control with Path Planning

I have worked on setting up the Robot, Servo commanding, calibration, robot Inverse-Kinematic and Dynamic control methods. Followed by applications of advanced controllers in Torque-Control domain like Robust control (Sliding-Mode) and concerning the uncertainties. Moreover, I used the Damped Least-Squares and Singular Value Decomposition (SVD) methods for avoiding the singularities.

key words: Delta parallel robot, Kinematic control, Dynamic control, SVD

IMU-Based Motion Tracking with Kalman Filter Drift Compensation

Designed and implemented a compact motion-tracking patch with Arduino Nano and a TDK MPU-9250 9-axis IMU sensor for a table-tennis racket monitoring. Acquisition of acceleration and orientation (Euler angle) data. Implemented a Kalman filter to suppress gyro drift and attitude-estimation stability.

key words: Sensor Fusion, Kalman Filter, Orientation Estimation

TEACHING & LABORATORY EXPERIENCE

Head of Teaching Assistants – Electrical Lab

Fall 2023 – Spring 2025

- Extensive hands-on experience with lab instruments: digital multimeter (DMM), signal generator, and oscilloscope; Frequency analysis (Bode Plots), low/high-pass RLC filters, and motor PWM control.
- Supervised and trained the team of TAs; Delivered weekly lectures, demonstrations and guided TAs through experiments. Ensured smooth lab operations across multiple sections.

Head of Teaching Assistants – Mechatronics I Lab

Fall 2025-Fall 2026

- Practical experience in Arduino-based implementation, pneumatic actuation; Frequency-domain analysis, including FFT-based signal inspection and audio/sound filtering. Expertise in embedded control, signal processing, and hardware–software interfacing within laboratory environments.
- Coordinating lab sections and mentoring TAs; leading demonstrations and troubleshooting Mechatronics systems. Managing lab logistics, and safety.

Mechatronics I Laboratory Development Committee

Spring 2025-Fall 2025

- Collaborating with faculty and fellow teaching assistants to design, direct, document, and implement new experimental modules for the Mechatronics I Laboratory.

RESEARCH ASSISTANT

The Human-Machine Interaction and Control (THinC) Lab, UNC Charlotte

- Conducting research funded by the National Science Foundation (NSF) on traffic networks to reduce the congestion time in highways using variable speed limits (VSL), for regulating the traffic in mixed environments where both Autonomous and Human-Driven vehicles co-exist.

Mechatronics Lab, Isfahan University

- Working on the development, modeling, and control of a 3-DOF Delta parallel; Implementing inverse-kinematic and dynamic control algorithms, including torque-based and velocity-based control strategies. Assisting in hardware integration and experimental trajectory-tracking controllers.

TECHNICAL SKILLS

- **Control, Estimation, Robotics:** Feedback Linearization, Model Predictive Control (MPC), Kalman and Extended Kalman Filters (EKF), PID, Lyapunov based, Linear systems.
- **Data Analysis and Signal Processing:** Frequency-domain analysis, FFT, Analog / digital filtering, sensor fusion, Lab Experience (Oscilloscope, DMM, Signal Generator, etc.).
- **ML & Data-Driven:** Gaussian Process (GP) modeling, Singular value Decomposition (SVD), Regression based ML.
- **Programming and Software:** MATLAB, Simulink, Python, C/C++, Arduino IDE, NI myDAQ, and Abaqus.

LANGUAGE

- **English** (Fluent)
- **German** B2
- **Persian** (Native)